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LAB EXERCISE 18- VOLUME IN KUBERNETES CLUSTER

VOLUMES IN KUBERNETES PROVIDE A WAY TO STORE AND SHARE DATA AMONG CONTAINERS IN A POD. THIS LAB EXERCISE WILL GUIDE YOU THROUGH THE CREATION AND USE OF DIFFERENT TYPES OF VOLUMES IN KUBERNETES.

STEP 1: SET UP KUBERNETES CLUSTER

ENSURE YOU HAVE ACCESS TO A KUBERNETES CLUSTER. YOU CAN USE A LOCAL SETUP WITH MINIKUBE, KIND, OR USE A CLOUD-BASED KUBERNETES SERVICE.

STEP 2: CREATE A POD WITH AN EMPTYDIR VOLUME

CREATE A FILE NAMED **EMPTYDIR-POD.YAML** WITH THE FOLLOWING CONTENT:

```
APIVERSION: V1
KIND: POD
METADATA:
  NAME: TEST-VOL1
SPEC:
  CONTAINERS:
    - IMAGE: NGINX
      NAME: TEST-CONTAINER
      VOLUMEMOUNTS:
        - MOUNTPATH: /DATA
          NAME: FIRST-VOLUME
  VOLUMES:
    - NAME: FIRST-VOLUME
      EMPTYDIR: {}
```

```

GNU nano 7.2
apiVersion: v1
kind: Pod
metadata:
  name: test-vol1
spec:
  containers:
  - image: nginx
    name: test-container
    volumeMounts:
    - mountPath: /data
      name: first-volume
  volumes:
  - name: first-volume
    emptyDir: {}

```

IN THIS MANIFEST:

THE `EMPTYDIR` VOLUME IS A TEMPORARY STORAGE THAT IS CREATED WHEN THE `POD` IS ASSIGNED TO A NODE AND EXISTS AS LONG AS THE `POD` IS RUNNING.

APPLY THE MANIFEST TO CREATE THE `POD`:

```
KUBECTL APPLY -F EMPTYDIR-POD.YAML
```

```

aml
pod/test-vol1 created

```

STEP 3: VERIFY THE `POD` AND VOLUME

CHECK THE STATUS OF THE `POD`:

```
KUBECTL GET PODS
```

NAME	READY	STATUS	RESTARTS	AGE
test-vol1	1/1	Running	0	15s

ACCESS THE `POD`'S SHELL:

```
KUBECTL EXEC -IT TEST-VOL1 -- BASH
```

INSIDE THE `POD`, CREATE A FILE IN THE `/DATA` DIRECTORY:

```
TOUCH F1 F2 F3
```

```

ash
root@test-vol1:/# touch f1 f2 f3

```

YOU SHOULD SEE THE TEXT "HELLO, KUBERNETES!".

EXIT THE `POD` SHELL:

```
EXIT
```

STEP 4: CREATE A `POD` WITH A `HOSTPATH` VOLUME

CREATE A FILE NAMED **HOSTPATH-POD.YAML** WITH THE FOLLOWING CONTENT:

```
APIVERSION: v1
KIND: POD
METADATA:
  NAME: TEST-VOL2
SPEC:
  CONTAINERS:
    - IMAGE: COOLGOURAV147/NGINX-CUSTOM
      NAME: TEST-CONTAINER
      VOLUMEMOUNTS:
        - MOUNTPATH: /DATA
          NAME: FIRST-VOLUME
  VOLUMES:
    - NAME: FIRST-VOLUME
      HOSTPATH:
        PATH: /TEST
```

```
GNU nano 7.2
apiVersion: v1
kind: Pod
metadata:
  name: test-vol2
spec:
  containers:
    - image: coolgourav147/nginx-custom
      name: test-container
      volumeMounts:
        - mountPath: /data
          name: first-volume
  volumes:
    - name: first-volume
      hostPath:
        path: /test
```

IN THIS MANIFEST:

THE HOSTPATH VOLUME MOUNTS A DIRECTORY FROM THE HOST INTO THE POD AT /MNT/DATA.

THE BUSYBOX CONTAINER WRITES THE CURRENT DATE TO A LOG FILE EVERY 5 SECONDS.

APPLY THE MANIFEST TO CREATE THE POD:

```
KUBECTL APPLY -F HOSTPATH-POD.YAML
```

```
aml
pod/test-vol2 created
```

STEP 3: VERIFY THE POD AND VOLUME

CHECK THE STATUS OF THE POD:

KUBECTL GET PODS

NAME	READY	STATUS	RESTARTS	AGE
test-vol1	1/1	Running	0	2m8s
test-vol2	0/1	ContainerCreating	0	11s

ACCESS THE POD'S LOGS TO SEE THE DATA BEING WRITTEN:

KUBECTL EXEC -IT TEST-VOL2 -- BASH

```
ash
root@test-vol2:/# |
```

YOU SHOULD SEE THE DATES BEING APPENDED TO THE LOG FILE.

CHECK THE HOST DIRECTORY TO VERIFY THE VOLUME:

IF YOU HAVE ACCESS TO THE NODE RUNNING THE POD, YOU CAN VERIFY THE DATA IN /TMP/DATA/LOG.TXT.